

Associating Real Objects with Virtual Models for VR Interaction

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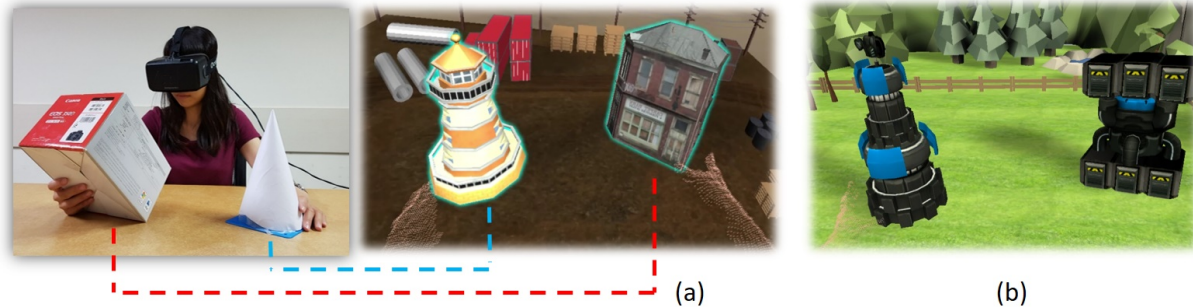


Figure 1: A VR user manipulating real objects. (a)The virtual view displayed on the head-mounted display. (b) Another view with a different virtual theme.

ABSTRACT

In this paper, we present a prototype system that is capable of associating real objects with virtual models and turning the table top into imaginary virtual scenes. A user can interact with these objects when she or he is immersed in the virtual environment. To accomplish this goal, a vision-based system is developed to online recognize and track the real objects in the scene. The corresponding virtual models are retrieved based on their tactile shapes. They are displayed and moved on a head-mounted display (HMD) according to tracked object poses. The experiment demonstrates that our prototype system can find reasonable association between real and virtual objects, and users are interested in the novel interaction.

CCS CONCEPTS

• **Computing methodologies** → **Virtual reality; Mixed / augmented reality; Tracking.**

KEYWORDS

3D interaction, tactile shape, object tracking

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1 INTRODUCTION

With the release of affordable equipment, virtual reality (VR) has become more and more popular and attractive for general consumers. Head-mounted Display (HMD) and 3D controllers or joysticks are the most common devices. Users can see the virtual world through HMD and interact with the virtual environment by controllers. Recently, users can even use their hands to interact with virtual objects through new devices incorporating hand tracking techniques. However, when users wear HMDs, they are unaware of physical environment. Present VR systems are usually set up in an empty room to prevent users from colliding with the surroundings. The above limitation urges us to consider whether we can make the physical surroundings perceptible and even interactive, when a user is immersed in the virtual environment.

To address these questions, this paper proposes finding objects in the real world and turning them into virtual ones. This way prevents users from unintended collision in real environment, and users can also gain passive haptic feedback from the supports of real props. To associate a physical object with a virtual one, we utilize computer vision techniques to recognize real objects in a scene and track their 3D movements. These objects can then be visualized in their virtual counterparts. Since the object shape of a virtual theme is often unordinary, it is not easy to find exact counterparts in appearance. Our work aims at finding counterparts with similar gripping feedback. Besides, we think the association between virtual and physical objects should be adaptable. Users are supposed to change their virtual scenes in the same physical room, and their physical surroundings can be adjusted. An interesting work proposed by Hettiarachchi and Wigdor [Hettiarachchi and Wigdor 2016] repurposed everyday objects and matched them to virtual models for augmented reality. We share similar concepts of object repurposing. Nevertheless, our matching technique considers not only the whole appearance shape but also tactile details, such as edges. Our system can provide more reasonable haptic feedback.

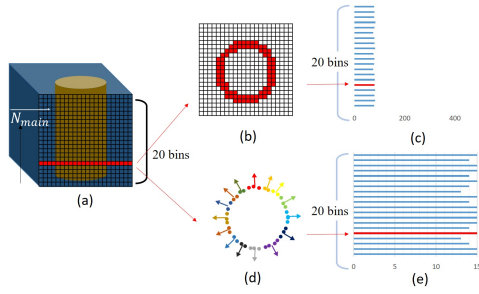


Figure 2: Examples of the feature extraction of Histogram of Surface Voxel (HSV) and Histogram of Surface Normal (HSN). (a) The voxel grids of an object. (b) The valid voxels in one of the 20 slices. Voxels with points are shown in red. (c) The illustration of HSV. (d) The illustration of the point clusters according to surface normals. (one color for one cluster). (e) The illustration of HSN.

2 OUR APPROACH

Our system is mainly divided into three stages. The first stage of *offline feature extraction* is to extract features from real and virtual models to alleviate run-time computation. We propose a matching scheme that considers the shape features of entire models and the tactile features measured by the surface normal cues. For the whole shapes, we first used an efficient shape matching method, Ensemble of Shape Function (ESF) [Wohlkinger and Vincze 2011]. The ESF descriptor is designed for shape recognition and provides reasonable matching results. We further incorporate a feature Histogram of Surface Voxel (HSV) to enhance the distinction in shape volume. Along the major axis of a shape $\vec{N_{main}}$, our system divides the space into 20 slices and calculates the voxel grid in each slice as shown in Fig. 2(b) and (c).

Even though the combination of ESF and HSV features can help us find objects in similar shapes and volume, the tactile feedback may not be so similar. For example, when a user sees a virtual object with a smooth surface but the real object she touched is with sharp edges and corners, they usually get confused and report the inconsistency. Therefore, we design an additional feature to amend this tactile issue. Histogram of Surface Normal (HSN) feature records the variations of surfaces perpendicular to the $\vec{N_{main}}$. We estimate the normal vector of each 3D sample point in a slice, and group the points of similar normal vectors (within a threshold) in to a cluster. A bin in HSN records the amount of clusters in that slice. Fig. 2(d) and (e) show the estimation of a bin in HSN vector.

In the second stage, we match the real objects with virtual ones based on the above three features. After finding the best pairs of virtual and real objects, our system then online maps the object information, extracted from a RGB-D camera mounted above the table, onto the virtual ones. We employed a RGB-D tracking method proposed in [Wu et al. 2016] for foreground extraction, object detection and object tracking in three dimensional space. The last stage is display, which shows up virtual models with a user-selected virtual scene on HMD. The movement of a virtual model in virtual environment is associated with the movement of its real counterpart. Also, a few visual aids are added to improve usability of the system as shown in Fig. 4.



Figure 3: Object pairing results with different feature combinations. The virtual model database contains various buildings. (a) Real models. (b) Pairing results with ESF features. (c) Pairing results by using the combination of ESF features and HSV features. (d) Pairing results by using the combination of ESF, HSV and HSN features.



Figure 4: Our system provides visual hints such as glowing contours of tangible objects and hand point clouds.

Please refer to our demo video to check the association of real objects with several virtual scenes. In the future work, we would like to expand our model datasets and conduct extensive user evaluation for the proposed system. Besides, we also intend to incorporate deep-learning-based methods for object tracking from RGB images such as [Huang et al. 2021] and analyze different features for tactile matching.

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